# MedAgent Clinical Decision Support

**Case ID:** case_1784a38312 | **Status:** APPROVE | **Physician:** Dr. Alex Vance | **Date:** 2026-10-15 20:11 UTC

**CRITICAL DISCLAIMER: Decision-support draft generated by AI agents. It is not a diagnosis and must be reviewed, corrected and approved by a licensed physician before any clinical use.**

## Patient Intake & Presentation

**Patient:** Age: 68, Sex: Male
**Notes & History:** 68yo male with 4-day history of productive cough with purulent rust-colored sputum, right-sided pleuritic chest pain, and high-grade fever (38.8°C). Dyspnea on mild exertion. Crackles audible at right lower lung base. No prior immunosuppression.

## Clinical Summary / Impression

Clinical evaluation indicates features consistent with Community-Acquired Pneumonia, characterized pathophysiologically by acute microbial infection triggering inflammatory exudation and leukocyte accumulation in pulmonary alveolar spaces, causing parenchymal consolidation and impaired oxygen diffusion. Key supporting clinical indicators include fever, cough, sputum, while secondary differential considerations include Acute Kidney Injury, Chronic Kidney Disease. Targeted diagnostic workup and guided serial clinical monitoring are recommended to establish definitive management.

## Ranked Differential Diagnoses

| # | Condition | Likelihood | Key Supporting Findings | Citations |
|---|---|---|---|---|
| 1 | **Community-Acquired Pneumonia** | **HIGH** | fever, cough, sputum, crackles, pleuritic | S2, S8 |
| 2 | **Acute Kidney Injury** | **MODERATE** | Renal Impairment | S_AcuteK_1, S_AcuteK_2 |
| 3 | **Chronic Kidney Disease** | **MODERATE** | Renal Impairment | S_Chroni_1, S_Chroni_2 |
| 4 | **Pulmonary Embolism** | **LOW** | pleuritic chest | S_Pulmon_1, S_Pulmon_2 |

## Laboratory Findings & Identified Patterns

| Analyte / Test | Result | Unit | Reference Interval | Flag |
|---|---|---|---|---|
| **C-Reactive Protein** | **128** | mg/L | 0 – 5 | **CRITICAL HIGH** |
| **WBC Count** | **16.2** | x10^3/uL | 4.5 – 11 | **HIGH** |
| **Neutrophils (%)** | **82** | % | 40 – 75 | **HIGH** |
| **Procalcitonin** | **1.4** | ng/mL | 0 – 0.1 | **HIGH** |
| **BUN** | **24** | mg/dL | 7 – 20 | **HIGH** |
| **Serum Creatinine** | **1.1** | mg/dL | 0.7 – 1.3 | NORMAL |

**Inferred Diagnoses from Laboratory Analysis:**
• **[SEVERE] Acute Bacterial Infection / Sepsis Risk** (Likelihood: HIGH): Leukocytosis (16.2 x10^3/uL) with left shift and markedly elevated inflammatory/sepsis biomarkers. *Mechanism:* Invasive pathogen eliciting acute bone marrow leukocyte demargination, bandemia, and systemic inflammatory endothelial activation with capillary leak.
• **[MODERATE] Prerenal Azotemia / Dehydration** (Likelihood: MODERATE): Elevated BUN-to-Creatinine ratio of 21.8:1 (BUN 24.0 mg/dL, Creatinine 1.10 mg/dL), characteristic of volume contraction or decreased cardiac output. *Mechanism:* Decreased effective renal perfusion pressure increasing proximal tubular reabsorption of urea relative to creatinine without intrinsic tubular structural necrosis.

**Synthesized Clinical Patterns:**
• **[SOON] Leukocytosis with Neutrophilia:** Elevated total leukocyte count with neutrophil predominance, suggesting active bacterial infection or severe acute inflammation.
• **[ROUTINE] Raised Inflammatory Markers:** Systemic inflammatory activation evidenced by elevated acute phase reactants.
• **[SOON] Renal Impairment:** Elevated nitrogenous wastes and/or reduced filtration rate indicating acute or chronic renal dysfunction.

## Suggested Diagnostic Workup

• Sputum gram stain and culture
• Blood cultures prior to antibiotics
• Follow-up chest radiograph in 6 weeks
• Renal ultrasound to exclude hydronephrosis

• Spot urine albumin-to-creatinine ratio
• CT Pulmonary Angiography (CTPA)

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**Physician Attestation:** I, Dr. Alex Vance, hereby confirm that I have reviewed, verified, and adjusted this decision-support synthesis. Decision: **APPROVE**.
**Reviewer Notes:** No specific notes recorded.
**Signature:** _____ **Date:** 2026-10-15 20:11 UTC